IRREDUCIBLE COMPONENTS OF A BASE CHANGE MAP ONTO IRREDUCIBLE COMPONENTS, AND THE MAP IS SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the surjectivity argument; no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-image-irreducible is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Base change along separable algebraic closures preserves irreducible components. Same chapter, Lemma lemma-separably-closed-field-irreducible-components, invoked at proof lines 1645-1648. Supplies the reduction to K equal to the separable closure of k.
- Generalizations lift along flat morphisms. Morphisms, Lemma morphisms-lemma-generalizations-lift-flat, invoked at proof lines 1655-1656. Supplies that generic points map to generic points.
- Integral morphisms are universally closed. Morphisms, Lemma morphisms-lemma-integral-universally-closed, invoked at proof lines 1659-1660. Supplies that the image of an irreducible component is closed.
- Inverse image of an irreducible component is a nonempty union of components. Same chapter, Lemma lemma-inverse-image-irreducible, invoked at proof lines 1665-1667. Supplies the surjectivity statement.

Source reference. Original lemma 04KW: sources/stacks-varieties/pinned-original-source.tex, source lines 1596–1596. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038H: sources/stacks-varieties/pinned-original-source.tex, source lines 1524–1524. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let K/k be an extension of fields. Let X be a scheme over k. For every irreducible component $\overline{T} \subset X_K$ the image of $\overline{T}$ in X is an irreducible component in X. This defines a canonical map*

$$IrredComp(X_K) \longrightarrow IrredComp(X)$$

which is surjective.

Proof. Consider the diagram

$$\begin{array}{ccc} X_K & \longleftarrow & X_{\overline{K}} \\ \downarrow & & \downarrow \\ X & \longleftarrow & X_{\overline{k}} \end{array}$$

where $\overline{K}$ is the separable algebraic closure of K, and where $\overline{k}$ is the separable algebraic closure of k. By Lemma 038H the morphism $X_{\overline{K}} \to X_{\overline{k}}$ induces a bijection between irreducible components. Hence it suffices to show the lemma for the morphisms $X_{\overline{k}} \to X$ and $X_{\overline{K}} \to X_K$. In other words we may assume that $K = \overline{k}$.

The morphism $p : X_{\overline{k}} \to X$ is integral, flat and surjective. Flatness implies that generalizations lift along p, see Morphisms, Lemma 03HV. Hence generic points of irreducible components of $X_{\overline{k}}$ map to generic points of irreducible components of X. Integrality implies that p is universally closed, see Morphisms, Lemma 01WM. Hence we conclude that the image $p(\overline{T})$ of an irreducible component is a closed irreducible subset which contains a generic point of an irreducible component of X, hence $p(\overline{T})$ is an irreducible component of X. This proves the first assertion. If $T \subset X$ is an irreducible component, then $p^{-1}(T) = T_K$ is a nonempty union of irreducible components, see Lemma 04KW. Each of these necessarily maps onto T by the first part. Hence the map is surjective. □